8.6

1

(11) $\int_0^2 \frac{dx}{(x^2-1)^{\frac{3}{2}}}$

$= \lim_{a\to 1^-} \int_0^a \frac{dx}{(x^2-1)^{\frac{3}{2}}} + \lim_{b\to 1^+} \int_b^2 \frac{dx}{(x^2-1)^{\frac{3}{2}}}$

$x = \sec\theta, \quad dx = \sec\theta\tan\theta\, d\theta$

$\lim_{a\to 1^-} \int_{\sec\theta=0}^{\sec\theta=a} \frac{\sec\theta}{\tan^2\theta} d\theta + \lim_{b\to 1^+} \int_{\sec\theta=b}^{\sec\theta=2} \frac{\sec\theta}{\tan^2\theta} d\theta$

$\frac{\sec\theta}{\tan^2\theta} \to \frac{C}{s^2} = \csc\cdot\cot$

$\lim_{a\to 1^-} [-\csc\theta]_{\sec\theta=0}^{\sec\theta=a} + \lim_{b\to 1^+} [-\csc\theta]_{\sec\theta=b}^{\sec\theta=2}$

$= \infty$

(16) $\int_0^{\pi} \csc x\, dx$

$= \left[\frac{1}{2} \ln \left| \frac{1-\cos x}{1+\cos x} \right| \right]_0^{\pi}$

$= \infty - \infty$ (발산)

(17) $\int_0^1 \frac{\sin^{-1} x}{\sqrt{1-x^2}} dx$ $\quad (\sin^{-1} x)' = \frac{1}{\sqrt{1-x^2}}$

$\sin^{-1} x = t$

$\int_0^{\frac{\pi}{2}} t\, dt = \left[\frac{1}{2} t^2 \right]_0^{\frac{\pi}{2}}$

$= \frac{\pi^2}{8}$ //

(19) $\int_1^4 \frac{dx}{\sqrt{x^2-1}}$ $\quad 1+\tan^2\theta = 16$

$x = \sec\theta$

$\int_0^{\cos\theta=\frac{1}{4}} \frac{\sec\theta\tan\theta}{\tan\theta} d\theta = \left[\ln|\sec\theta + \tan\theta| \right]_0^{\cos\theta=\frac{1}{4}}$

$= \ln(4+\sqrt{15})$ //

2

(2) $\int_0^1 x^p \ln x \, dx$

$(p > -1)$ 인 경우

$$= \lim_{t \to 0^+} \int_t^1 x^p \ln x \, dx$$

$$= \lim_{t \to 0^+} \left(\left[\ln x \cdot \frac{1}{p+1} x^{p+1} \right]_t^1 - \int_t^1 \frac{1}{p+1} x^p \, dx \right)$$

$$= \lim_{t \to 0^+} - \ln t \cdot \frac{1}{p+1} t^{p+1} - \left[\frac{1}{(p+1)^2} x^{p+1} \right]_t^1$$

$$= \lim_{t \to 0^+} - \ln t \cdot \frac{1}{p+1} t^{p+1} - \frac{1}{(p+1)^2} + \frac{1}{(p+1)^2} t^{p+1}$$

$$= -\frac{1}{(p+1)^2} \quad (\text{수렴}) \quad //$$

$(p \leq -1$ 인 경우$)$

$$\int_0^1 x^p \ln x \, dx$$

$$= \left[\ln x \cdot \frac{1}{p+1} x^{p+1} \right]_0^1 - \int_0^1 \frac{1}{p+1} x^p \, dx$$

$$= -\infty + C \quad (\text{발산})$$